Ashley Li

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Education

University of Illinois Urbana-Champaign — Champaign, IL
BS Computer Science and Linguistics | GPA: 4.0/4.0 — *Aug 2024 – May 2028 (Expected)*

- Relevant coursework: Systems Programming, Intro to Algorithms and Models of Computation, Computer Networks, Database Systems, Data Structures, Computer Architecture, Linear Algebra, Statistics and Probability
- In progress: Distributed Systems, Corpus Linguistics

Work Experience

Microsoft — Redmond, WA
Explorer Intern, Software Engineering and Program Management — *May 2026 – Aug 2026*

- Designed and integrated diagnostic signals to Copilot Voice, surfacing device, network, and audio issues to users in real time
- Architected and built diagnostic event plumbing between WebRTC backend and Voice; implemented reusable and extensible components with TypeScript and React
- Authored product requirements and design documents; drove collaboration with engineering, PM, design, and partner teams
- Identified and fixed a production regression that prevented WebRTC errors from being logged in Voice telemetry

University of Illinois — Champaign, IL
Undergraduate Research Assistant, Professor Lawrence Angrave — *Aug 2024 – Jul 2026*

- Doubled the precision of lecture scene detection, a CV video-segmentation algorithm, by improving the detection of gradual transitions like scrolling and handwriting; drove technical direction across four semesters
- Evaluated CV algorithms in Python against a hand-labeled 30,000 frame dataset, deployed to production and staging for thousands of engineering students
- Awarded $8,000 Computing Research Association REU grant for the 2025–2026 academic year
- First-authored a work-in-progress paper published in ASEE IL/IN Regional Conference

University of Washington Robinson Center Summer Stretch — Seattle, WA
Teacher's Aide — *Jul 2025*

- TA for accelerated Survey of Linguistics and Algebra 1 classes for middle-school students
- Collaborated with instructors on grading and classroom management, facilitated discussion, and provided individual academic support

Leadership

UIUC Association for Computing Machinery — Champaign, IL
Infrastructure Co-Chair — *Sep 2025 – Present*

- Run infrastructure platform for event management, merchandise sales, member services, and more; resolve production incidents and feature requests
- Developing USD bank-transfer feature (Stripe, DynamoDB) to cap corporate sponsorship fees at $5
- Support 80+ person executive council; lead student project teams
- Using AWS (DynamoDB, Lambda), Stripe, Microsoft Entra ID, Cloudflare, Node.js, TypeScript, Git

Projects

Distributed Key-Value Store | *Rust, Tokio, Tonic, gRPC*

- Implementing distributed key-value store based on Cassandra supporting linearizability, eventual consistency, and read repair.
- Implemented distributed log querier and gossip-based membership protocol with failure detection.

Personal Website | *React, Tailwind CSS, GithubPages*

- Lightweight static website built with React Router and Tailwind CSS. Deployed with GithubPages.

Class Website | *Next.js, Tailwind CSS, Supabase (Postgres), Auth.js (NextAuth.js), Vercel*

- Website for students to suggest and vote on class activities and sign up for volunteer opportunities.
- Developed with Next.js App Router and Auth.js Microsoft authentication, deployed with Vercel. Managed relational data, authentication state, and real-time updates with Supabase.
- Implemented role-based access control, allowing admin to publish, edit, and deny student submissions.

Technical Skills

Languages: Rust, C++, C, TypeScript, Python, Java, MIPS Assembly, SQL (Postgres), JavaScript, HTML/CSS, LaTeX
Frameworks & Libraries: AWS, React, Node.js, OpenCV, NumPy, Ollama, Next.js, Tailwind, Auth.js
Developer Tools: Git, Docker, VS Code, Valgrind, Jupyter Notebook